SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

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Code of Conduct
I Jaswanth S (192525214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Assigned Problem Statement No. 33

AI-Powered Smart Municipal Asset Management and Maintenance Platform

A Literature Survey Report

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1. Introduction

Municipal corporations are custodians of large, heterogeneous portfolios of public assets, including roads, parks, streetlights, drainage systems, public buildings, and water infrastructure. The performance of these assets directly determines the quality of civic services delivered to citizens. Historically, asset management in municipal bodies has relied on manual inspection registers, paper-based work orders, and reactive repair cycles, which are slow, inconsistent, and costly to sustain as urban infrastructure ages and expands.

Artificial Intelligence (AI), the Internet of Things (IoT), Geographic Information Systems (GIS), and cloud computing have collectively opened new possibilities for transforming static, fragmented asset records into a live, predictive, and centrally managed digital ecosystem. Condition-monitoring sensors, computer-vision inspection tools, and machine-learning based failure-prediction models allow municipal engineers to move from a "fix-when-broken" paradigm to a "fix-before-failure" paradigm, improving reliability while lowering long-term maintenance expenditure.

This report presents a literature survey conducted for the assigned problem statement, "AI-Powered Smart Municipal Asset Management and Maintenance Platform." It reviews ten research contributions relevant to digital asset inventories, IoT-based condition monitoring, GIS-based decision support, and AI/ML-driven predictive maintenance, compares their methodologies and outcomes, identifies limitations and research gaps in the existing body of work, and proposes future research directions aligned with the assigned problem.

1.1 Problem Statement

Municipal assets often deteriorate due to delayed maintenance and fragmented management systems. Manual inspections make it difficult to prioritize repairs effectively. An AI-powered municipal asset management platform should maintain a digital inventory of public assets, monitor asset conditions using IoT sensors, predict maintenance requirements, prioritize repair schedules, and generate infrastructure performance reports, with the overall goal of improving service quality while reducing maintenance costs.

1.2 Objectives of the Proposed Platform

1. Digitize municipal asset records.
2. Monitor asset conditions continuously.
3. Predict maintenance requirements.
4. Prioritize repair schedules.
5. Integrate GIS-based asset mapping.
6. Generate maintenance analytics.
7. Improve infrastructure utilization.
8. Enhance public service delivery.

1.3 Expected Outcomes

1. Improved infrastructure reliability.
2. Reduced maintenance costs.
3. Better public asset utilization.
4. Faster maintenance planning.

5. Enhanced municipal services.
6. Data-driven infrastructure management.
7. Increased operational efficiency.
8. Improved citizen satisfaction.

1.4 Literature Survey Methodology

The literature search was carried out across reputed academic sources, including IEEE Xplore, Springer Link, ScienceDirect, MDPI, ASCE Library, and Google Scholar-indexed repositories such as ResearchGate and Semantic Scholar. Search terms combined the core themes of the problem statement — for example, "AI predictive maintenance municipal infrastructure," "GIS asset management," "IoT road condition monitoring," and "digital twin pavement management" — to capture work spanning digital inventories, condition monitoring, predictive analytics, and spatial decision support.

Papers were shortlisted based on three inclusion criteria: (i) direct relevance to municipal, civic, or public-infrastructure asset management rather than purely industrial or manufacturing contexts; (ii) use of a defined methodology, dataset, or case study rather than purely promotional or vendor material; and (iii) availability in a peer-reviewed journal, conference proceeding, or credible scholarly preprint. Ten papers meeting these criteria, spanning publication years 2002 to 2026 and covering GIS, IoT, machine learning, and Digital Twin approaches, were selected for detailed review and comparison in Sections 2 and 3.

2. Literature Survey

Ten research works spanning GIS-based asset management, IoT/AI predictive maintenance, computer-vision road monitoring, and digital-twin frameworks were selected from peer-reviewed journals, conference proceedings, and reputed preprint/review sources. Each paper is summarized below in terms of its objective, methodology, tools/dataset used, and principal findings.

2.1 GIS-Integrated Intelligent System for Roads and Bridges (Salim et al., 2002) [1]

This early but foundational study combined a GIS platform (ArcView/MapObjects) with a heuristic, rule-based AI expert-system shell to optimize winter road and bridge maintenance, using snow removal in Black Hawk County, Iowa, as a case study. The system integrated the Iowa Department of Transportation's road inventory with expert-encoded maintenance rules, demonstrating that GIS and AI together can improve resource allocation for infrastructure upkeep. However, the reasoning engine was static and non-adaptive, limiting its applicability to a single maintenance activity rather than a comprehensive multi-asset platform.

2.2 Integrated Risk-Assessment Framework for Municipal Infrastructure (Shahata & Zayed, 2016) [2]

Shahata and Zayed developed a risk-based decision-support model for prioritizing the repair and renewal of road, water, and sewer (RWS) networks. Risk indices were derived using a mixed Delphi–Analytical Hierarchy Process approach across eighteen sub-factors and then aggregated using K-means clustering into a single integrated risk index. Applied to real data from the City of Guelph, the model found that pipe/road size and accessibility were the dominant risk drivers. The framework, while effective for planning, depends on expert-elicited weightings rather than continuously updated sensor data, and does not incorporate real-time IoT monitoring or self-learning prediction.

2.3 Road Condition Monitoring Using Smart Sensing and AI (Ranyal, Sadhu & Jain, 2022) [3]

This systematic review surveyed road-condition-monitoring literature published between 2017 and 2022, covering vision-based AI methods and next-generation sensing technologies such as smartphones, drones, vehicle-mounted RGB/thermal cameras, lasers, and ground-penetrating radar. The review found substantial progress in automated distress detection and classification but highlighted persistent challenges in quantifying distress severity and density, and noted the absence of standardized, cross-region benchmark datasets for training generalizable models.

2.4 Smartphone-Based Crowdsourced Pavement Diagnostics (Staniek, 2021) [4]

Staniek investigated smartphone-based crowdsourcing for pavement-condition diagnostics, using accelerometer, gyroscope, and GPS data classified through C4.5 decision trees, support vector machines, and naive Bayes classifiers. The C4.5 algorithm achieved detection accuracy as high as 98.6% when combining accelerometer and gyroscope inputs. While cost-effective and scalable, the approach is sensitive to device mounting position and vehicle type, and the crowdsourced data stream is not natively linked to a centralized municipal asset registry.

2.5 GIS-Based Solutions for Public Asset Management (Popov et al., 2025) [5]

This state-of-the-art review examined how GIS technologies support public building and infrastructure asset management, including data visualization, cost assessment, risk assessment, and maintenance-strategy selection.

The authors concluded that although GIS is a common integrating element across asset types, current tools remain fragmented, and a unified framework that couples GIS with AI-based predictive analytics is still largely absent from practice.

2.6 AI-Driven Predictive Maintenance Using Ensemble Learning (Bin Masud et al., 2025) [6]

This study compared K-Nearest Neighbors, Support Vector Classifier, Random Forest, and XGBoost algorithms for predicting equipment failure from operational metrics such as temperature, torque, rotational speed, and tool wear. XGBoost consistently outperformed the other classifiers in predictive accuracy. The evaluation, however, was conducted on a synthetic, generic industrial dataset rather than real municipal-asset data, and the study did not address model interpretability, which is important for accountable public-sector decision-making.

2.7 Digital Twin and Deep Learning for Pavement Condition (Gurrola-Mijares et al., 2025) [7]

This paper proposed a formalized highway Digital Twin architecture composed of a Physical Space, a Data Interconnection layer, and a Virtual Space, and applied a novel parallel Gated Recurrent Unit (GRU) network to classify road-surface anomalies from low-cost Inertial Measurement Unit (IMU) data. The proposed model achieved 89.5% accuracy and an F1-score of 0.875, outperforming conventional classifiers. The framework was validated on a single highway case study, leaving its scalability to an entire municipal road network, and its integration with other asset classes, unproven.

2.8 Integrated BIM–GIS Framework for Bridge Asset Management (Anhiem, 2026) [8]

This recent work proposed a bidirectional data-exchange architecture between IFC-compliant Building Information Modelling (BIM) models and georeferenced GIS databases, together with mathematical formulations for condition-index computation, deterioration prediction, and maintenance prioritization. The framework was validated on an inventory of 47 bridges along the Juba–Nimule highway corridor. While effective for bridge-specific decision-making, the framework has not been tested on other municipal asset classes such as drainage systems, parks, or public buildings.

2.9 Multimodal IoT–AI Framework for Urban Facility Risk Assessment (2026) [9]

This study addressed fragmented multi-source municipal data by proposing a multimodal AI framework that fuses IoT sensor streams, facility images, inspection records, and complaint work-order text to assess urban-facility risk and prioritize response. The framework demonstrated improved lead time in failure prediction and more consistent risk ranking than manual, experience-based prioritization, but integration across heterogeneous legacy municipal IT/OT systems remains a significant implementation barrier.

2.10 Predictive Maintenance for Smart City Infrastructure — Review (2025) [10]

This review article surveyed the transition from reactive/scheduled maintenance to AI- and IoT-enabled predictive maintenance across bridges, roads, energy grids, and water systems. It synthesized real-world case studies to show gains in efficiency, resilience, and asset longevity, while identifying high upfront sensor-deployment costs and limited technical capacity as major adoption barriers for resource-constrained municipalities, particularly in developing regions.

3. Comparative Literature Survey Table

Table 1 consolidates the reviewed literature across seven parameters — author/year, research problem, methodology, dataset/tools, key findings, and limitations/research gap — to enable a side-by-side comparison of existing approaches to AI-powered municipal asset management.

#	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations / Research Gap
1	Salim, Strauss & Emch (2002)	Optimizing road & bridge maintenance scheduling	GIS (ArcView/MapObjects) integrated with a heuristic AI expert-system shell	Iowa DOT road/bridge inventory; Black Hawk County, Iowa case study	GIS–AI integration improved snow-removal asset allocation decisions	Rule-based, non-adaptive AI; single-purpose (snow removal); dated tool stack not generalizable to multi-asset platforms
2	Shahata & Zayed (2016)	Prioritizing repair/renewal of road, water and sewer networks	Delphi–Analytical Hierarchy Process (AHP) risk model with K-means clustering	Municipal RWS network data, City of Guelph case study	Produced an integrated risk index; pipe/road size and accessibility were dominant risk factors	Relies on expert-elicited weights; no real-time IoT sensing or predictive learning component
3	Ranyal, Sadhu & Jain (2022)	Automated road condition monitoring	Systematic review of vision-based AI and smart-sensor techniques (2017–2022)	Smartphones, drones, vehicle-mounted RGB/thermal cameras, GPR, LiDAR	AI and non-contact sensing significantly improve distress detection and classification	No standardized benchmark dataset; limited severity/density quantification; poor cross-region generalization
4	Staniek (2021)	Crowdsourced pavement condition diagnostics	Smartphone accelerometer/gyroscope/GPS data classified using C4.5, SVM and naive Bayes	Crowdsourced smartphone sensor data from moving vehicles	C4.5 decision tree achieved up to 98.6% detection accuracy for pavement anomalies	Data quality depends on device mounting and vehicle type; no linkage to a central asset registry
5	Popov, Mozaffari, RazaviAlavi & Jalaei (2025)	Managing public building and infrastructure assets	Systematic literature review of GIS applications in public asset management	Peer-reviewed literature corpus (state-of-the-art review)	GIS strengthens visualization, cost assessment, and maintenance-strategy selection	Review identifies fragmented tools; lacks a unified GIS + AI predictive-maintenance framework
6	Bin Masud et al. (2025)	Predicting equipment/asset failure proactively	Comparative ML: KNN, SVC, Random Forest, and XGBoost classifiers	Synthetic predictive-maintenance dataset (temperature, torque,	XGBoost delivered the highest fault-prediction accuracy among tested models	Synthetic, non-municipal dataset; limited model interpretability/explainability

#	Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations / Research Gap
				speed, tool wear)		
7	Gurrola-Mijares et al. (2025)	Continuous pavement condition assessment	Digital Twin framework (Physical–Data–Virtual layers) with a parallel GRU deep-learning network	Low-cost IMU sensor data from a highway case study	Achieved 89.5% accuracy and 0.875 F1-score, outperforming traditional classifiers	Validated on a single highway; scalability to multi-asset municipal networks untested
8	Anhiem (2026)	Managing national road-bridge inventories	Integrated BIM–GIS framework with IFC-to-GIS schema mapping and deterioration models	Inventory of 47 bridges, Juba–Nimule highway corridor	Enabled structured condition-index computation and maintenance prioritization	Small, region-specific sample; excludes non-bridge asset classes such as drainage or buildings
9	Smart Infrastructure IoT–AI Review (2026)	Fragmented data and manual risk prioritization in urban facility management	Multimodal AI framework combining IoT sensor streams with image/text inspection data	Inspection records, complaint work orders, facility images	Real-time analytics reduced downtime and improved failure-prediction lead time	Integration complexity across heterogeneous legacy municipal IT/OT systems
10	Predictive Maintenance for Smart-City Infrastructure Review (2025)	Transition from reactive to predictive city-asset maintenance	Conceptual review of AI/IoT-based predictive-maintenance architectures and case studies	Cross-sectoral case studies (bridges, roads, energy grids, water systems)	Predictive strategies reduce disruptive repairs and extend asset service life	High upfront sensor/infrastructure cost; barriers to adoption in resource-constrained municipalities

Table 1: Comparative summary of reviewed literature on AI-powered municipal asset management and maintenance.

4. Critical Analysis and Research Gaps

A synthesis of the reviewed literature reveals several recurring strengths and weaknesses across existing approaches to municipal asset management.

4.1 Strengths Observed in Existing Work

- GIS platforms provide a mature and widely adopted foundation for spatially organizing and visualizing municipal asset data [1], [2], [5], [8].
- Machine-learning and deep-learning classifiers (XGBoost, GRU, C4.5, SVM) demonstrate high accuracy in detecting specific failure modes and pavement anomalies from sensor or image data [3], [4], [6], [7].
- Digital Twin and BIM–GIS architectures show that structured, layered system designs can support condition-index computation and deterioration forecasting [7], [8].
- Risk-based decision models (Delphi–AHP, K-means clustering) offer a principled way to prioritize limited repair budgets across multiple asset types [2].

4.2 Limitations and Gaps in Existing Research

- Asset-class fragmentation: Most studies focus on a single asset type (roads, bridges, or a single facility) rather than an integrated platform covering roads, drainage, streetlights, parks, and buildings simultaneously [1], [7], [8].
- Limited real-world, municipal-scale validation: Several ML models are trained and evaluated on synthetic or narrow case-study datasets rather than large, heterogeneous municipal asset inventories [6], [7], [8].
- Weak integration between GIS and predictive AI: GIS-centric studies rarely incorporate continuously updated IoT sensor streams or self-learning prediction models, while AI-centric studies rarely provide spatial asset-mapping capability [2], [5].
- Interoperability and legacy-system barriers: Municipal IT/OT ecosystems are heterogeneous, and multimodal frameworks report significant integration difficulty when combining sensor data, images, and textual work-order records [9].
- Model interpretability and accountability: Ensemble and deep-learning models achieve strong accuracy but offer limited explainability, which is a concern for publicly accountable repair-prioritization decisions [6].
- Cost and scalability barriers: High upfront investment in sensors and platforms restricts adoption in smaller or resource-constrained municipal corporations [10].

Overall, the literature confirms that the individual building blocks required for a smart municipal asset management platform — digital inventories, IoT condition monitoring, GIS mapping, and AI-based prediction — are each reasonably well studied in isolation. The principal research gap is the absence of a single, unified, interoperable, and explainable platform that integrates all of these components across the full range of municipal asset classes while remaining affordable for typical municipal corporations.

4.3 Mapping of Assigned Objectives to Literature Coverage

Table 2 maps the eight objectives defined in the assigned problem statement to the extent of coverage found across the reviewed literature, highlighting which objectives are well addressed and which remain comparatively under-researched.

Objective	Coverage in Literature	Supporting References
1. Digitize municipal asset records	Partial — addressed mainly through GIS/BIM inventories for specific asset types, not a unified multi-asset registry	[1], [5], [8]
2. Monitor asset conditions continuously	Strong — IoT and smart-sensor based monitoring is extensively studied	[3], [4], [7], [9]
3. Predict maintenance requirements	Strong — multiple ML/DL predictive models validated	[6], [7], [10]
4. Prioritize repair schedules	Moderate — risk-based prioritization models exist but rely on expert weighting	[2], [9]
5. Integrate GIS-based asset mapping	Strong — GIS is the most mature component across the literature	[1], [2], [5], [8]
6. Generate maintenance analytics	Moderate — analytics demonstrated at case-study scale, not city-wide	[6], [7], [9]
7. Improve infrastructure utilization	Limited — mostly inferred as an outcome rather than directly measured	[2], [10]
8. Enhance public service delivery	Limited — citizen-facing impact rarely quantified in technical studies	[9], [10]

Table 2: Coverage of assigned platform objectives across the reviewed literature.

The mapping shows that condition monitoring and predictive maintenance (Objectives 2 and 3) are the most extensively researched areas, largely driven by advances in IoT sensing and machine learning. In contrast, holistic digitization of multi-asset records (Objective 1) and citizen-facing service-delivery improvements (Objective 8) receive comparatively less direct attention in the technical literature, reinforcing the research gap identified above and motivating the future directions proposed in Section 5.

5. Future Research Directions

- Develop a unified, multi-asset digital inventory schema that links GIS spatial layers with IoT sensor identifiers across roads, drainage, streetlights, buildings, and parks within a single data model.
- Design lightweight, explainable AI/ML models (e.g., interpretable ensemble or attention-based methods) for predictive maintenance that municipal engineers can audit and trust for repair-prioritization decisions.
- Investigate low-cost sensing and crowdsourcing strategies (smartphone-based, citizen-reporting) as economical alternatives or supplements to dedicated IoT sensor networks in budget-constrained municipalities.
- Extend Digital Twin and BIM–GIS architectures, validated so far mainly on bridges and highways, to drainage networks, public buildings, and streetlighting systems.
- Build standardized, open benchmark datasets covering diverse municipal asset types and climatic/regional conditions to improve model generalizability and enable fair comparison across studies.
- Design middleware or API-based interoperability layers to bridge legacy municipal IT/OT systems with modern AI/IoT analytics platforms, reducing integration overhead.
- Incorporate citizen-facing dashboards and complaint/work-order text analytics alongside sensor data to align predictive maintenance with actual service-quality perception.

6. Conclusion

This literature survey reviewed ten research contributions relevant to the design of an AI-powered smart municipal asset management and maintenance platform. The reviewed works collectively demonstrate that GIS, IoT sensing, computer vision, machine learning, and Digital Twin technologies each provide valuable, individually validated capabilities for improving infrastructure reliability and maintenance efficiency. However, the survey identifies a clear research gap: no single framework in the reviewed literature integrates digital multi-asset inventories, continuous IoT-based condition monitoring, GIS-based spatial decision support, and explainable AI-driven predictive maintenance into one interoperable, cost-effective platform suitable for typical municipal corporations. Addressing this gap, along with the interoperability, explainability, and affordability challenges identified above, represents a promising direction for future research and for the design of the proposed platform.

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